

Period 3 elements: Physical Properties

element	structure & bonding	melting point	electrical conductivity
Na	➤ giant metallic lattice, strong metallic bonds ➤ factors affecting metallic bond strength: - no. of delocalised valence electrons - charge density of cation $\propto \frac{q^+}{r^+}$	➤ melting point of Al > Mg > Na because metallic bond strength of ____ > ____ > ____	➤ conducts electricity in molten state due to presence of sea of delocalised electrons ➤ electrical conductivity of Al > Mg > Na because no. of delocalised valence electrons for ____ > ____ > ____
Mg			
Al			
Si	➤ giant molecular structure, strong covalent bonds between atoms	➤ very high	➤ poor conductor of electricity, metalloid
P ₄	➤ simple molecular structure, weak id-id between molecules ➤ strength of id-id interactions for ____ > ____ > ____ ➤ factor affecting strength of id-id: no. of electrons, size & polarisability of electron cloud	➤ melting point of ____ > ____ > ____ because strength of id-id interactions between molecules for ____ > ____ > ____	➤ does not conduct electricity due to absence of mobile charge carriers
S ₈			
Cl ₂			

Period 3 Oxides: Physical & Chemical Properties

compound	structure & bonding	melting point/ °C	electrical conductivity	reaction with water	pH
Na ₂ O	giant ionic lattice, strong ionic bonds	high	conducts electricity in molten state due to presence of free moving ions	Na ₂ O(s) + H ₂ O(l) → 2NaOH(aq)	
MgO				MgO(s) + H ₂ O(l) ⇌ Mg(OH) ₂ (s) ⇌ Mg(OH) ₂ (aq)	
Al ₂ O ₃				does not dissolve in water	
SiO ₂	giant molecular structure, strong covalent bonds between atoms	high	does not conduct electricity due to absence of mobile charge carriers	does not dissolve in water	
P ₄ O ₁₀	simple molecular structure, weak id-id between molecules	low		P ₄ O ₁₀ (s) + 6H ₂ O(l) → 4H ₃ PO ₄ (aq)	
SO ₃		m.p. of P ₄ O ₁₀ > SO ₃		SO ₃ (g) + H ₂ O(l) → H ₂ SO ₄ (aq)	

compound	ionic/ covalent	acid-base nature	reaction with HCl(aq) AND / OR NaOH(aq)
Na ₂ O	ionic	basic	Na ₂ O(s) + 2HCl(aq) → 2NaCl(aq) + H ₂ O(l)
NaOH		basic	NaOH(aq) + HCl(aq) → NaCl(aq) + H ₂ O(l)
MgO		basic	MgO(s) + 2HCl(aq) → MgCl ₂ (aq) + H ₂ O(l)
Mg(OH) ₂		basic	Mg(OH) ₂ (s) + 2HCl(aq) → MgCl ₂ (aq) + 2H ₂ O(l)
Al ₂ O ₃	ionic with covalent character	amphoteric	Al ₂ O ₃ (s) + 6HCl(aq) → 2AlCl ₃ (aq) + 3H ₂ O(l) Al ₂ O ₃ (s) + 2NaOH(aq) + 3H ₂ O(l) → 2 _____(aq)
Al(OH) ₃		amphoteric	Al(OH) ₃ (s) + 3HCl(aq) → AlCl ₃ (aq) + 3H ₂ O(l) Al(OH) ₃ (s) + NaOH(aq) → _____(aq)
SiO ₂	covalent	acidic	SiO ₂ (s) + 2NaOH(conc) → _____(aq) + H ₂ O(l)
P ₄ O ₁₀		acidic	P ₄ O ₁₀ (s) + 12NaOH(aq) → 4Na ₃ PO ₄ (aq) + 6H ₂ O(l)
SO ₃		acidic	SO ₃ (g) + 2NaOH(aq) → Na ₂ SO ₄ (aq) + H ₂ O(l)

Period 3 Chlorides: Physical & Chemical Properties

compound	Structure & bonding	melting point	electrical conductivity
NaCl	giant ionic lattice, strong ionic bonds	➤ high due to strong ionic bonds	conducts electricity in molten state due to presence of free moving ions
MgCl ₂			
AlCl ₃	simple molecular structure, weak id-id between molecules	➤ generally low due to weak id-id interaction between molecules	does not conduct electricity due to absence of mobile charge carriers
SiCl ₄		➤ melting point of _____ > _____ > _____ due to stronger id-id	
PCl ₅		➤ AlCl ₃ exists as a dimer: Al ₂ Cl ₆ , which only dissociates at high temp.	

compound	ionic/ covalent	effect on addition to water	reaction with water	pH
NaCl	ionic	dissolves readily	Hydration: $\text{NaCl(s)} + \text{aq} \rightarrow \text{Na}^{\text{+}}(\text{aq}) + \text{Cl}^{-}(\text{aq})$	7
MgCl ₂	ionic with covalent character	dissolves readily with slight hydrolysis	Hydration: $\text{MgCl}_2(\text{s}) + 6\text{H}_2\text{O(l)} \rightarrow [\text{Mg}(\text{H}_2\text{O})_6]^{2+}(\text{aq}) + 2\text{Cl}^{-}(\text{aq})$ Hydrolysis: $[\text{Mg}(\text{H}_2\text{O})_6]^{2+}(\text{aq}) \rightleftharpoons [\text{Mg}(\text{H}_2\text{O})_5(\text{OH})]^{+}(\text{aq}) + \text{H}^{+}(\text{aq})$ OR $[\text{Mg}(\text{H}_2\text{O})_6]^{2+}(\text{aq}) + \text{H}_2\text{O(l)} \rightleftharpoons [\text{Mg}(\text{H}_2\text{O})_5(\text{OH})]^{+}(\text{aq}) + \text{H}_3\text{O}^{+}(\text{aq})$	6.5
AlCl ₃	covalent	dissolves readily with appreciable hydrolysis	With limited amount of water: $\text{AlCl}_3(\text{s}) + 3\text{H}_2\text{O(l)} \rightarrow \text{Al}(\text{OH})_3(\text{s}) + 3\text{HCl(g)}$	
			With excess water: Hydration: $\text{AlCl}_3(\text{s}) + 6\text{H}_2\text{O(l)} \rightarrow [\text{Al}(\text{H}_2\text{O})_6]^{3+}(\text{aq}) + 3\text{Cl}^{-}(\text{aq})$ Hydrolysis: $[\text{Al}(\text{H}_2\text{O})_6]^{3+}(\text{aq}) \rightleftharpoons \text{___\$	

Hydrolysis

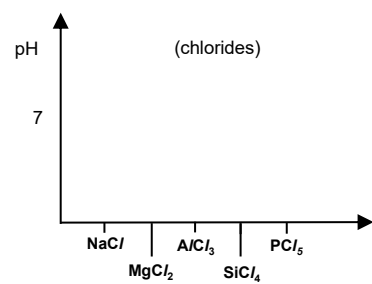
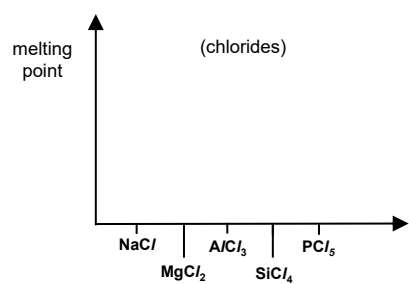
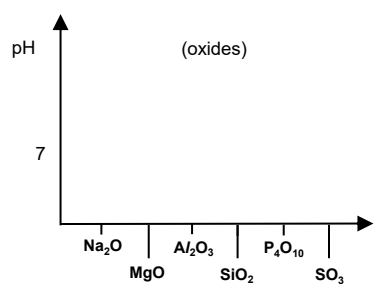
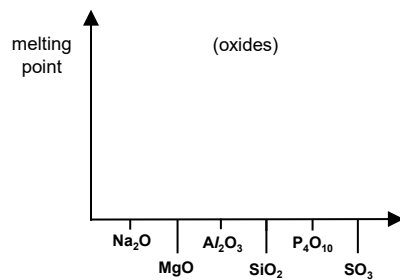
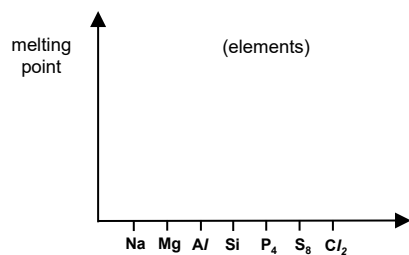
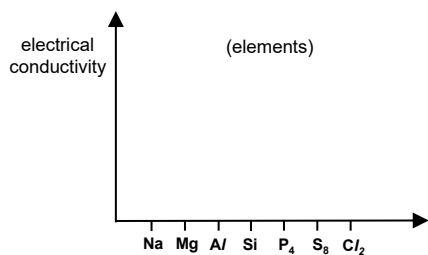
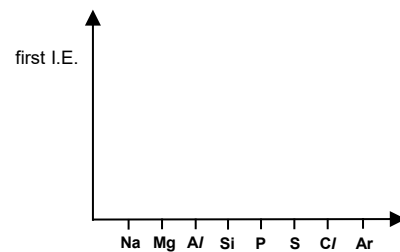
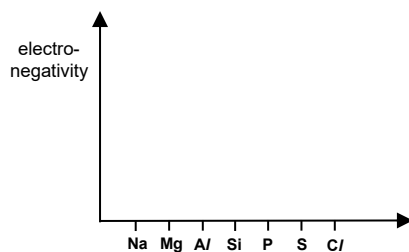
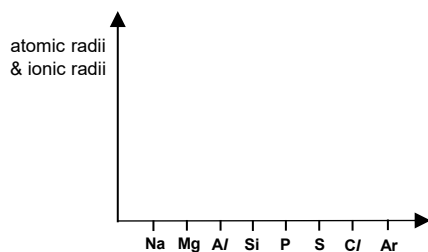
Note:

- Hydrolysis is a broadly defined term used to describe reactions that involves H₂O and bond breaking.
- Not all hydrolysis reactions have the same mechanism

- Explain why [Al(H₂O)₆]³⁺ hydrolyses to a larger extent than [Mg(H₂O)₆]²⁺
 - due to higher charge and smaller ionic radii, Al³⁺ has a _____ charge density and hence _____ polarising power
 - Al³⁺ _____ H₂O molecule to a larger extent
 - O—H bond in H₂O is _____ to a larger extent
 - O—H bond undergo heterolytic fission more readily, releasing H⁺

- Explain why CCl₄ is unable to hydrolyse unlike SiCl₄
 - C is a smaller atom, hence the four large Cl atoms _____ the approach of H₂O nucleophile from attacking the C atom
 - C does not have _____ orbitals that can accept lone pair of electrons from H₂O nucleophile

Graph sketches



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Na	➤ giant metallic lattice, strong metallic bonds ➤ factors affecting metallic bond strength: - no. of delocalised valence electrons - charge density of cation $\propto \frac{q^+}{r^+}$	➤ melting point of $Al > Mg > Na$ because metallic bond strength of <u>Al</u> > <u>Mg</u> > <u>Na</u>	➤ conducts electricity in molten state due to presence of sea of delocalised electrons ➤ electrical conductivity of $Al > Mg > Na$ because no. of delocalised valence electrons for <u>Al</u> > <u>Mg</u> > <u>Na</u>
Mg			
Al			
Si	➤ giant molecular structure, strong covalent bonds between atoms	➤ very high	➤ poor conductor of electricity, metalloid
P ₄	➤ simple molecular structure, weak id-id between molecules ➤ strength of id-id interactions for <u>S₈</u> > <u>P₄</u> > <u>Cl₂</u> ➤ factor affecting strength of id-id: no. of electrons, size & polarisability of electron cloud	➤ melting point of <u>S₈</u> > <u>P₄</u> > <u>Cl₂</u> because strength of id-id interactions between molecules for <u>S₈</u> > <u>P₄</u> > <u>Cl₂</u>	➤ does not conduct electricity due to absence of mobile charge carriers
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Na ₂ O	giant ionic lattice, strong ionic bonds	high	conducts electricity in molten state due to presence of free moving ions	$Na_2O(s) + H_2O(l) \rightarrow 2NaOH(aq)$	13-14
MgO				$MgO(s) + H_2O(l) \rightleftharpoons Mg(OH)_2(s) \rightleftharpoons Mg(OH)_2(aq)$	9-10
Al ₂ O ₃				does not dissolve in water	7
SiO ₂	giant molecular structure, strong covalent bonds between atoms	high	does not conduct electricity due to absence of mobile charge carriers	does not dissolve in water	7
P ₄ O ₁₀	simple molecular structure, weak id-id between molecules	low		$P_4O_{10}(s) + 6H_2O(l) \rightarrow 4H_3PO_4(aq)$	2
SO ₃		m.p. of P ₄ O ₁₀ > SO ₃		$SO_3(g) + H_2O(l) \rightarrow H_2SO_4(aq)$	2

compound	ionic/ covalent	acid-base nature	reaction with HCl(aq) AND / OR NaOH(aq)
Na ₂ O	ionic	basic	$Na_2O(s) + 2HCl(aq) \rightarrow 2NaCl(aq) + H_2O(l)$
NaOH		basic	$NaOH(aq) + HCl(aq) \rightarrow NaCl(aq) + H_2O(l)$
MgO		basic	$MgO(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2O(l)$
Mg(OH) ₂		basic	$Mg(OH)_2(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + 2H_2O(l)$
Al ₂ O ₃	ionic with covalent character	amphoteric	$Al_2O_3(s) + 6HCl(aq) \rightarrow 2AlCl_3(aq) + 3H_2O(l)$ $Al_2O_3(s) + 2NaOH(aq) + 3H_2O(l) \rightarrow 2 \text{ Na}^+[Al(OH)_4]^- (aq)$
Al(OH) ₃		amphoteric	$Al(OH)_3(s) + 3HCl(aq) \rightarrow AlCl_3(aq) + 3H_2O(l)$ $Al(OH)_3(s) + NaOH(aq) \rightarrow \text{ Na}^+[Al(OH)_4]^- (aq)$
SiO ₂	covalent	acidic	$SiO_2(s) + 2NaOH(conc) \rightarrow \text{ Na}_2SiO_3 (aq) + H_2O(l)$
P ₄ O ₁₀		acidic	$P_4O_{10}(s) + 12NaOH(aq) \rightarrow 4Na_3PO_4(aq) + 6H_2O(l)$
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MgCl ₂	ionic with covalent character	dissolves readily with slight hydrolysis	Hydration: $\text{MgCl}_2(\text{s}) + 6\text{H}_2\text{O(l)} \rightarrow [\text{Mg}(\text{H}_2\text{O})_6]^{2+}(\text{aq}) + 2\text{Cl}^-(\text{aq})$ Hydrolysis: $[\text{Mg}(\text{H}_2\text{O})_6]^{2+}(\text{aq}) \rightleftharpoons [\text{Mg}(\text{H}_2\text{O})_5(\text{OH})]^+(\text{aq}) + \text{H}^+(\text{aq})$ OR $[\text{Mg}(\text{H}_2\text{O})_6]^{2+}(\text{aq}) + \text{H}_2\text{O(l)} \rightleftharpoons [\text{Mg}(\text{H}_2\text{O})_5(\text{OH})]^+(\text{aq}) + \text{H}_3\text{O}^+(\text{aq})$	6.5
AlCl ₃	covalent	dissolves readily with appreciable hydrolysis	With limited amount of water: $\text{AlCl}_3(\text{s}) + 3\text{H}_2\text{O(l)} \rightarrow \text{Al(OH)}_3(\text{s}) + 3\text{HCl(g)}$	
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SiCl ₄	covalent	hydrolyses in water to produce white fumes of HCl(g), which dissolves in water to produce HCl(aq)	With limited amount of water: $\text{SiCl}_4(\text{l}) + 2\text{H}_2\text{O(l)} \rightarrow \text{SiO}_2(\text{s}) + 4\text{HCl(g)}$	
			With excess water: $\text{SiCl}_4(\text{l}) + 2\text{H}_2\text{O(l)} \rightarrow \text{SiO}_2(\text{s}) + 4\text{HCl(aq)}$	2
PCl ₅			With limited amount of water or when the water is cold: $\text{PCl}_5(\text{s}) + \text{H}_2\text{O(l)} \rightarrow \text{POCl}_3(\text{l}) + 2\text{HCl(g)}$ When more water is added: $\text{POCl}_3(\text{l}) + 3\text{H}_2\text{O(l)} \rightarrow \text{H}_3\text{PO}_4(\text{aq}) + 3\text{HCl(aq)}$	
			With excess water and water is not cold: $\text{PCl}_5(\text{s}) + 4\text{H}_2\text{O(l)} \rightarrow \text{H}_3\text{PO}_4(\text{aq}) + 5\text{HCl(aq)}$	2

Hydrolysis

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 - due to higher charge and smaller ionic radii, Al³⁺ has a higher charge density and hence higher polarising power
 - Al³⁺ polarises H₂O molecule to a larger extent
 - O—H bond in H₂O is weakened to a larger extent
 - O—H bond undergo heterolytic fission more readily, releasing H⁺

- Explain why CCl₄ is unable to hydrolyse unlike SiCl₄
 - C is a smaller atom, hence the four large Cl atoms hinders the approach of H₂O nucleophile from attacking the C atom
 - C does not have low-lying, vacant orbitals that can accept lone pair of electrons from H₂O nucleophile

Graph sketches

